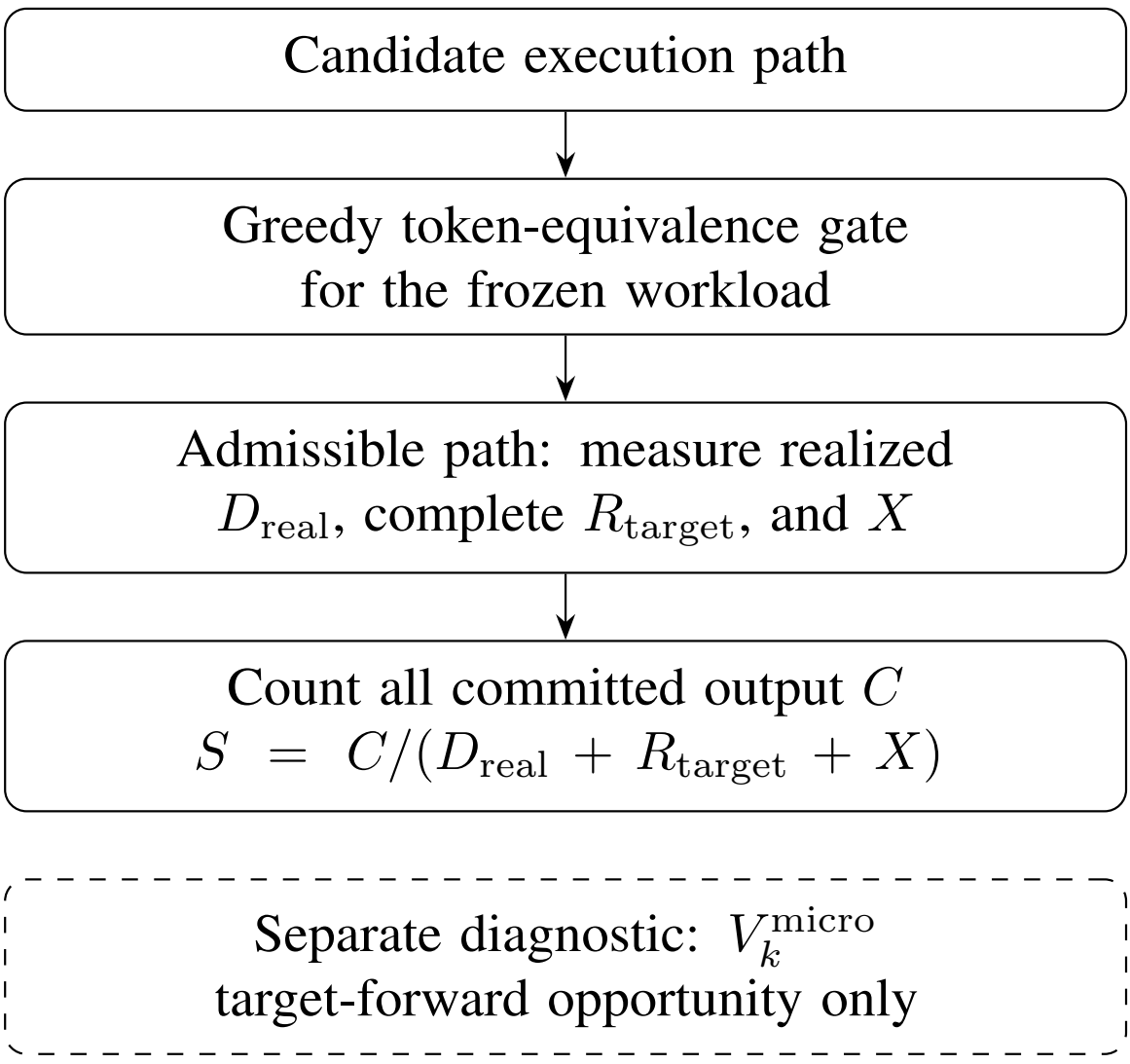
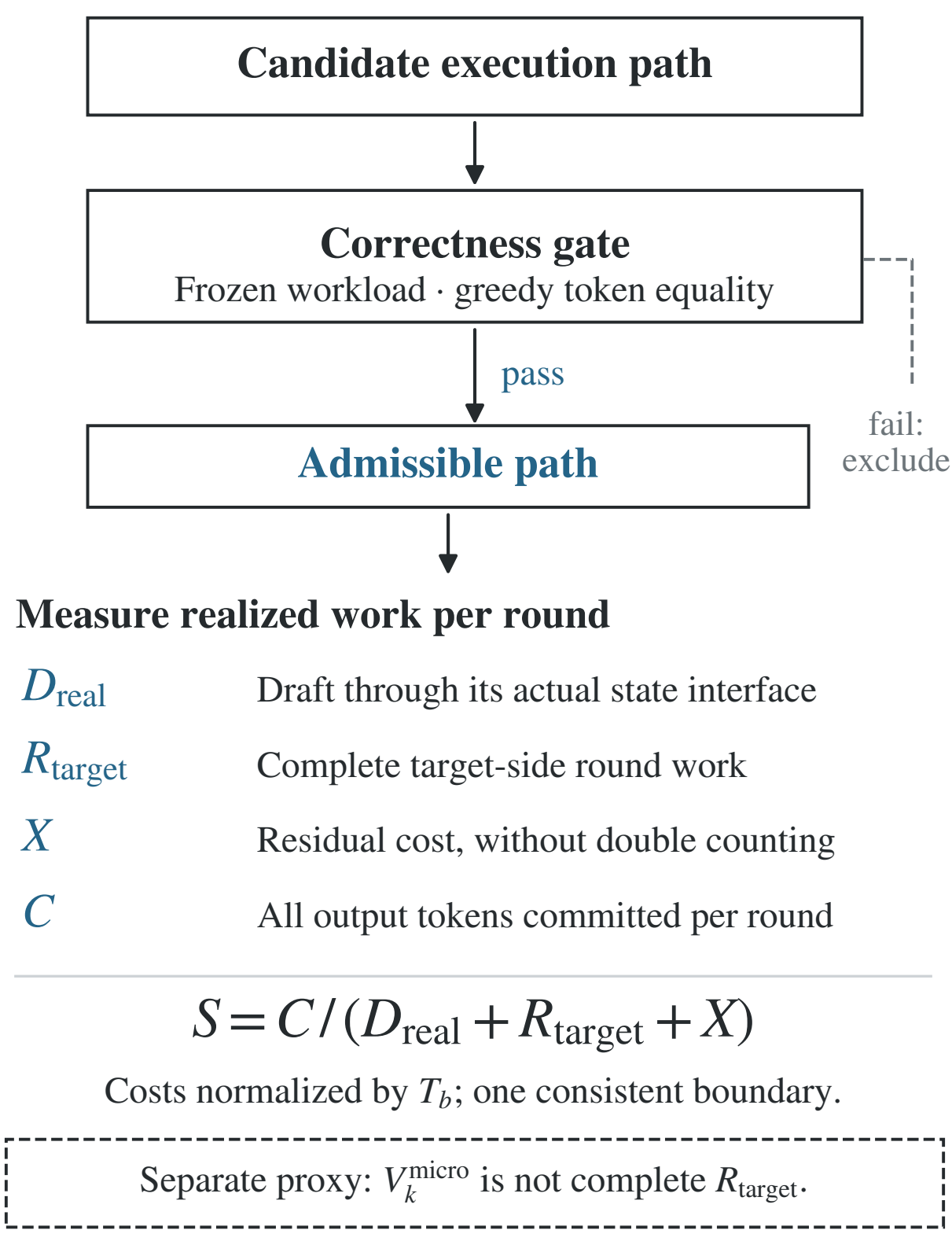
Figure 1: OLD / PRIOR candidate v0.2

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 2.077x)



PRIOR candidate v0.2

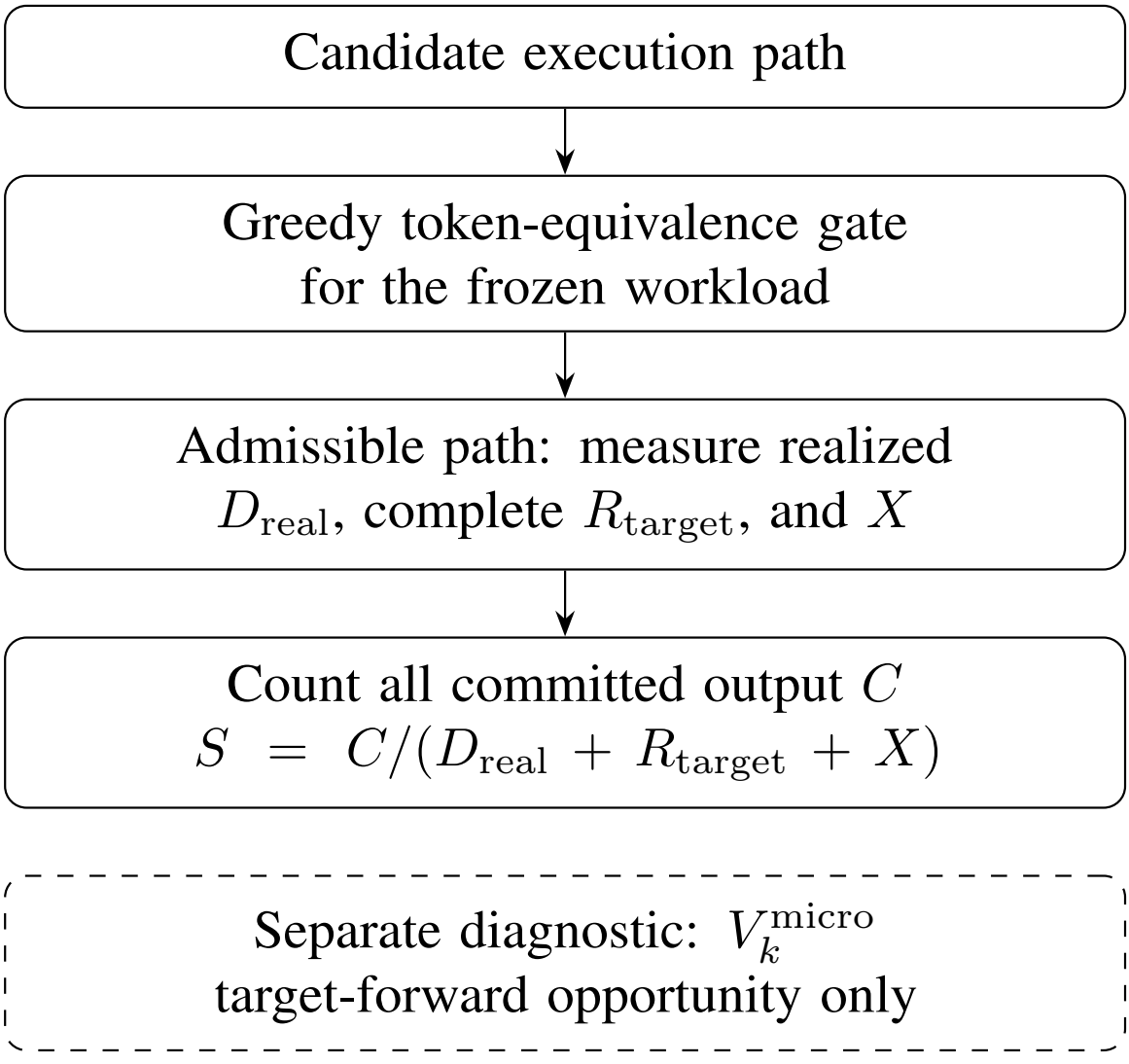


Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

Figure 1: OLD / SELECTED v0.3

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 2.077x)



SELECTED v0.3

Local opportunity does not ensure end-to-end gain

Observed case	Favorable local measurement		Realized outcome
M4 · HumanEval30 Preregistered, fixed k = 2	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + profitable** 1.387× · 90/90 exact
M4 · representative 24Q Frozen representative study	$V_2^{\text{micro}} = 1.022–1.076$ Shared Metal path; separate micro study	→	● **Exact + negative** 0.842× · 72/72 exact
RTX 5070 Ti Laptop CUDA batched verification	$V_2^{\text{micro}} = 0.983–1.009$ Near one target-decode step	→	□ **Correctness blocked** No admissible speedup
358H · NPU draft Qualified Phase-3 path	48.08 vs 30.81 tokens/s NPU vs CPU · isolated, 8,192 tokens	→	● **Exact + negative** 0.352× · 90/90 exact

Categorical cases, not a hardware ranking. Filled = qualified; open = blocked. Details and CIs: Figures 2–4.

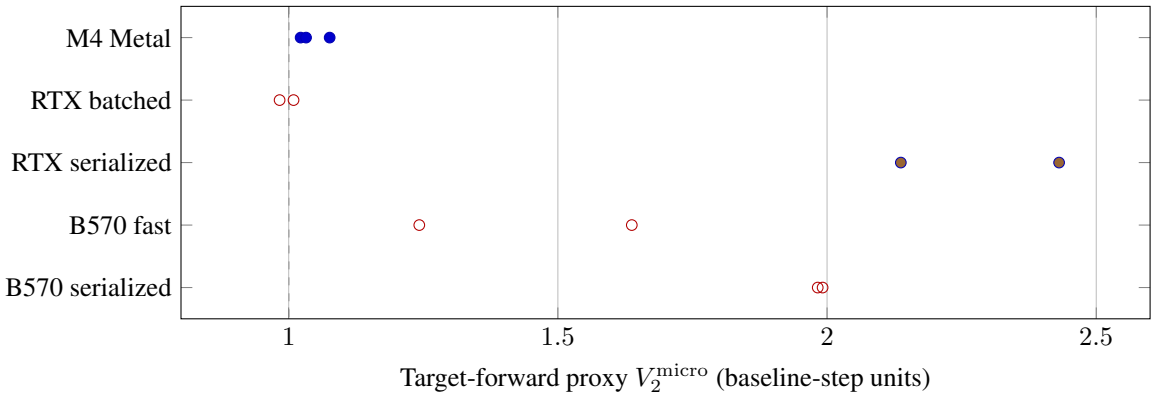
Micro cost ≠ complete target work; isolated throughput ≠ stateful cost. Speedups cover post-prefill decode.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

Figure 2: OLD / PRIOR candidate v0.2

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 6.75 in; scale 1.000x)



PRIOR candidate v0.2

Execution path

M4 Metal
prefix 512 / 2,048 / 8,192

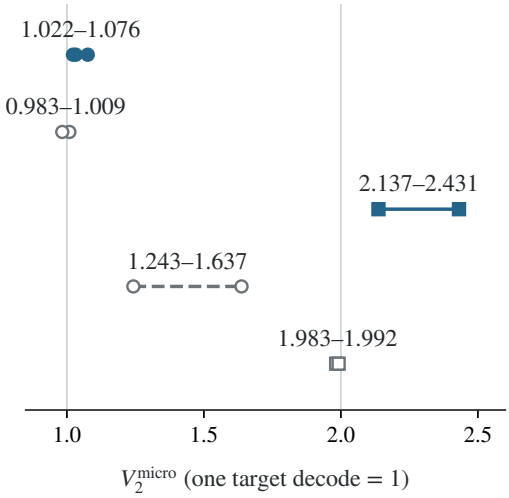
RTX 5070 Ti Laptop · batched
CUDA · ubatch = 2 · depth 0 / 256

RTX 5070 Ti Laptop · serialized
CUDA · ubatch = 1 · depth 0 / 256

B570 Vulkan · fast
Vulkan · context 512 / 2,048

B570 Vulkan · serialized
Vulkan · ubatch = 1 · 512 / 2,048

Measured target-forward cost



Correctness qualification

Qualified
Exact in tested cells

Blocked
Frozen workload gate failed

Qualified control
Exact secondary control

Diagnostic · blocked
45/90 pairs exact

Diagnostic · blocked
60/90 exact; still inadmissible

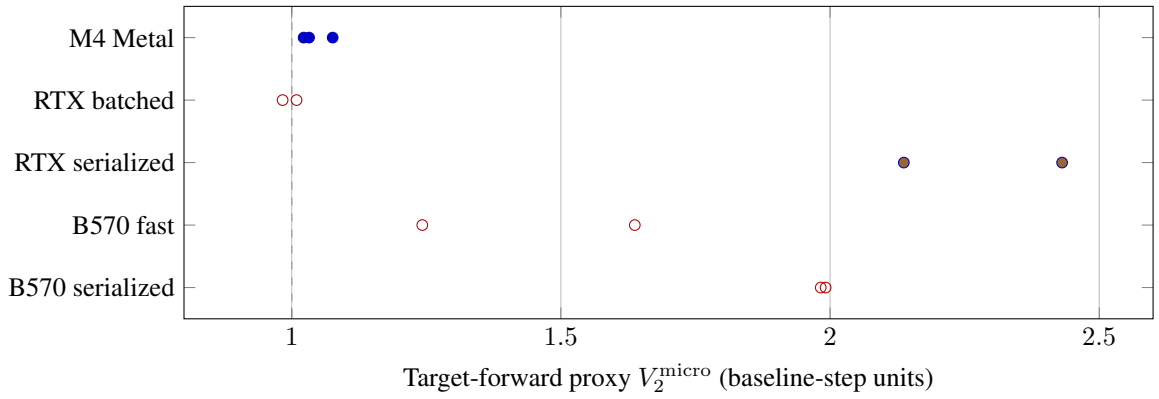
Filled = qualified; open = blocked/diagnostic. Circles = batched/fast; squares = serialized.
Ranges span tested contexts (not CIs); contexts are not fully matched. $V_2^{\mathrm{micro}} \neq R_{\mathrm{target}}$.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

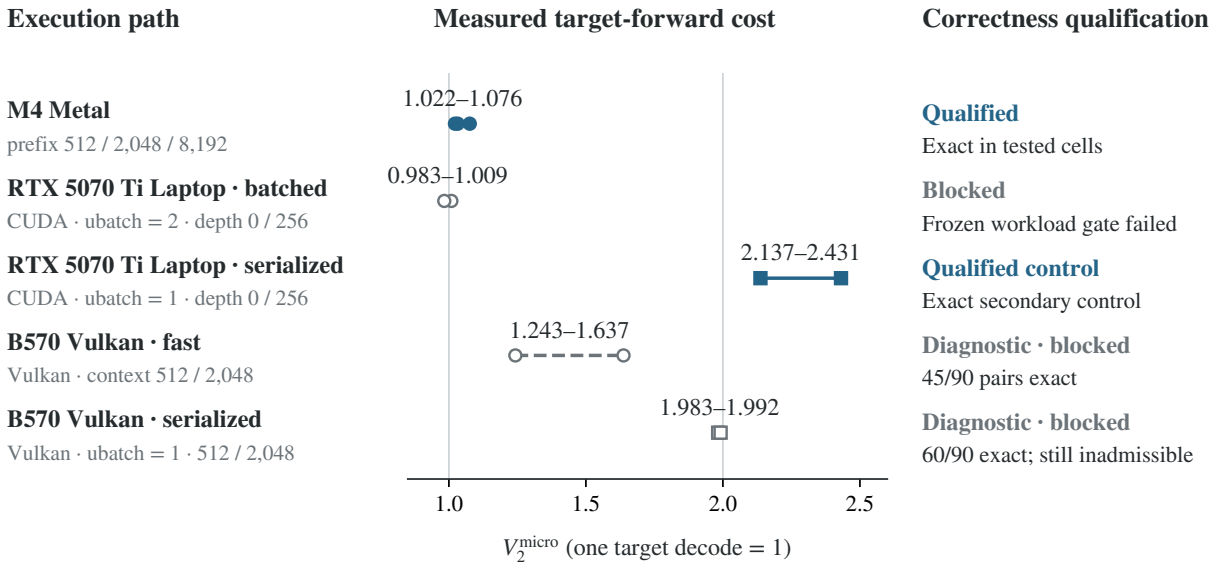
Figure 2: OLD / SELECTED v0.3

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 6.75 in; scale 1.000x)



SELECTED v0.3



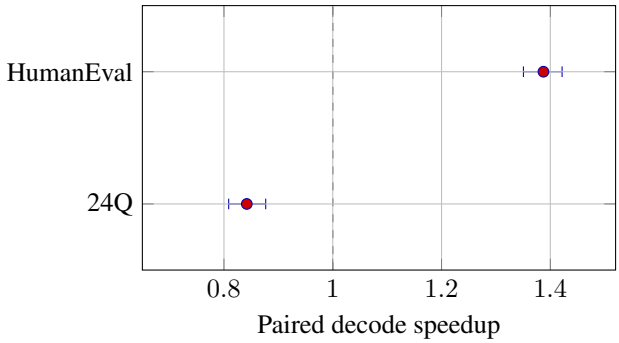
Filled = qualified; open = blocked/diagnostic. Circles = batched/fast; squares = serialized.
Ranges span tested contexts (not CIs); contexts are not fully matched. $V_2^{\mathrm{micro}} \neq R_{\mathrm{target}}$.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

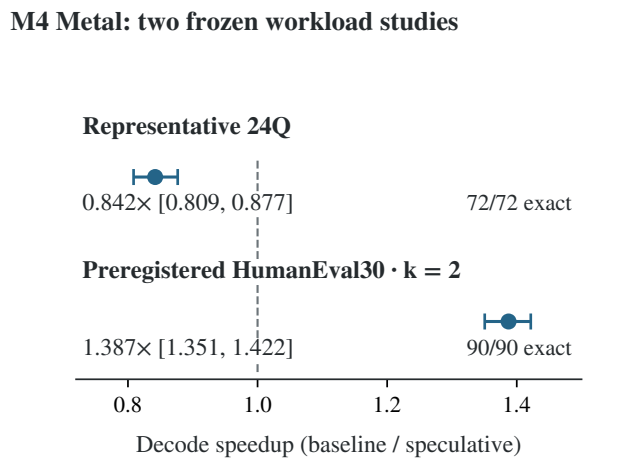
Figure 3: OLD / PRIOR candidate v0.2

Both panels 3.25 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 1.000x)



PRIOR candidate v0.2



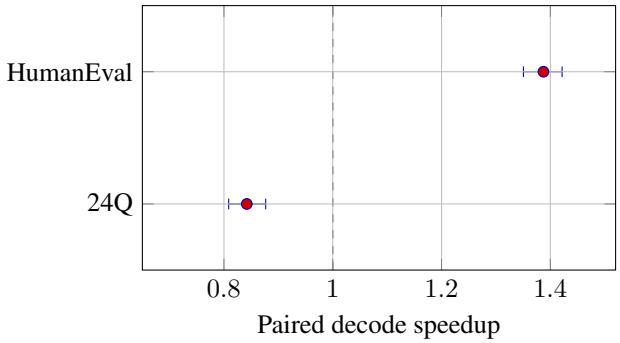
95% prompt-bootstrap CIs; separate workload studies.
Exact = token equality, not functional HumanEval pass@k.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

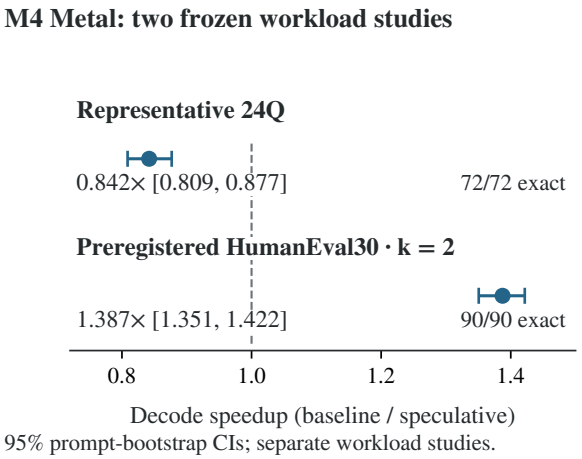
Figure 3: OLD / SELECTED v0.3

Both panels 3.25 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 1.000x)



SELECTED v0.3

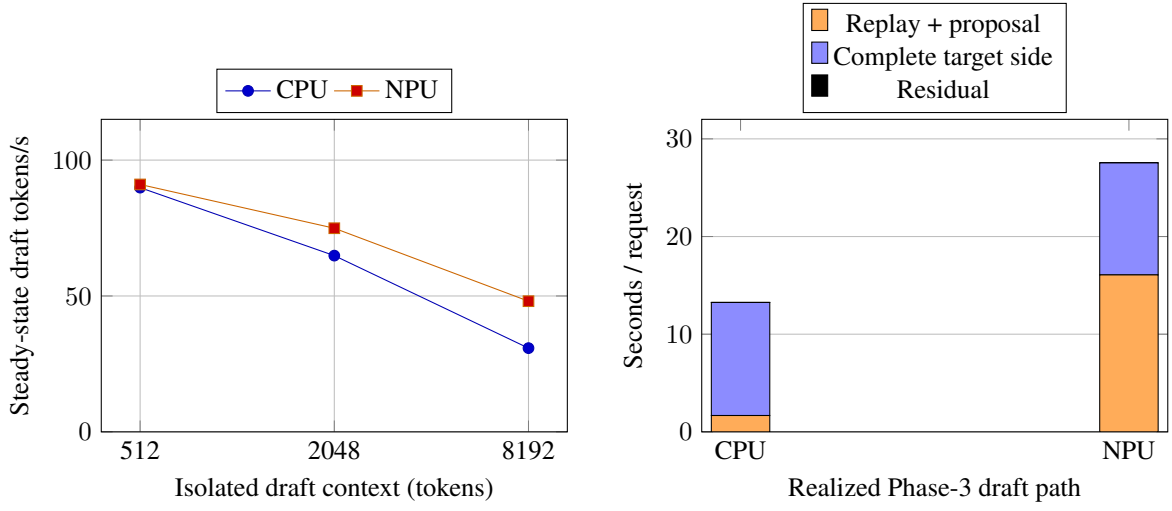


Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

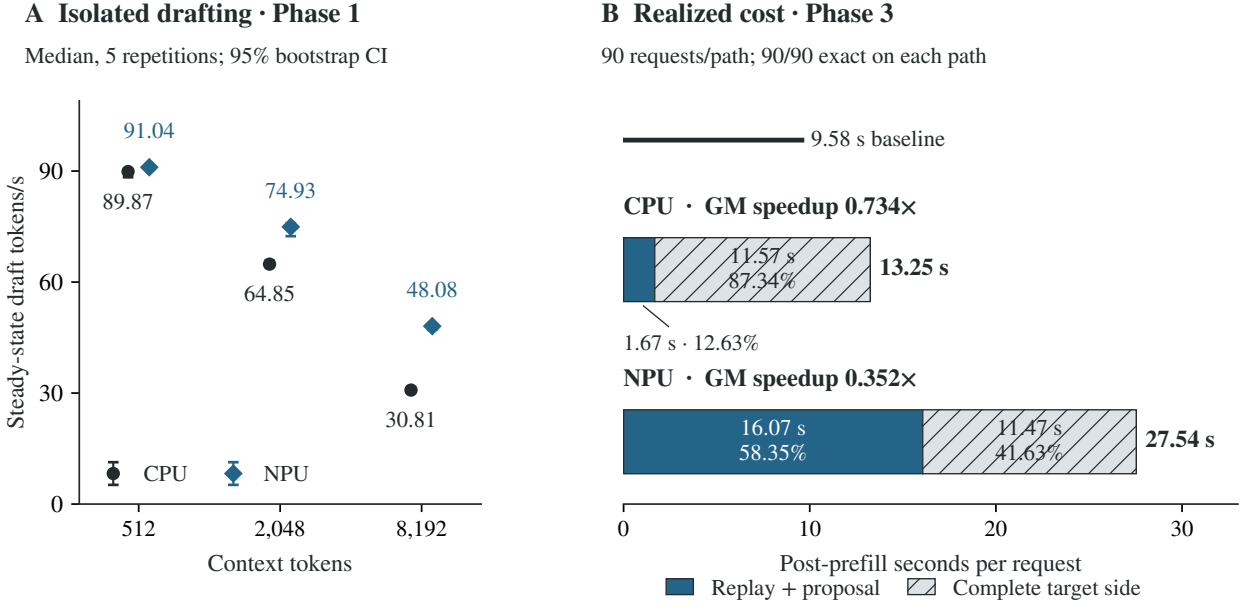
Figure 4: OLD / PRIOR candidate v0.2

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 6.75 in; scale 1.000x)



PRIOR candidate v0.2



Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.

Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.

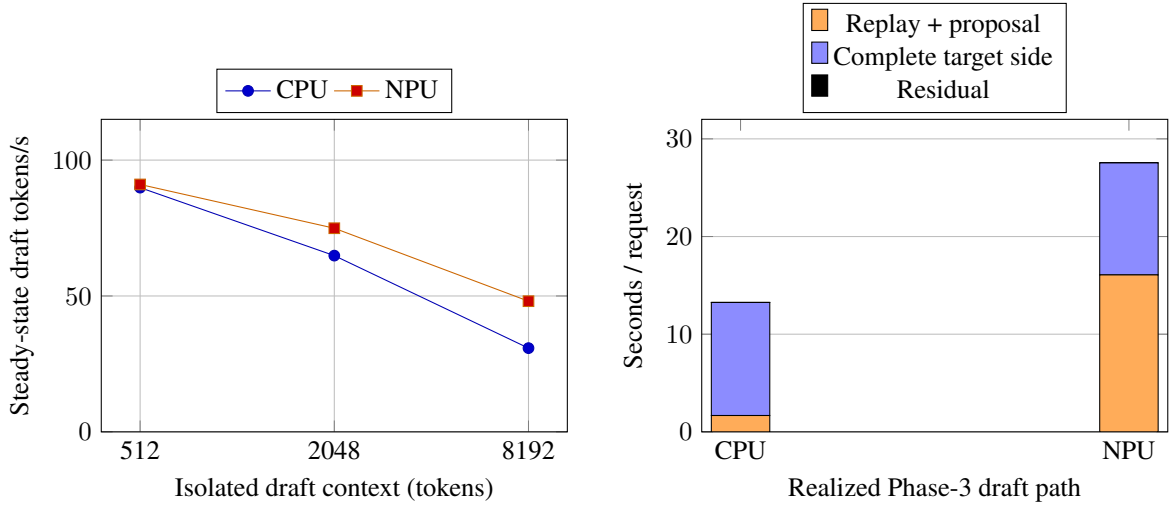
Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

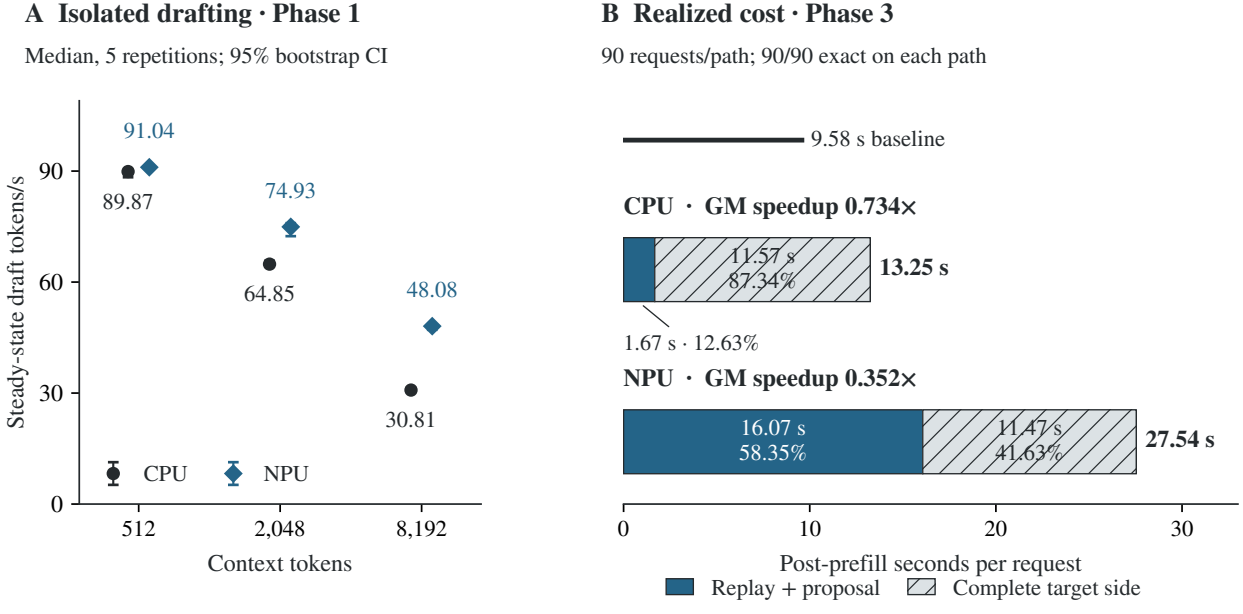
Figure 4: OLD / SELECTED v0.3

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 6.75 in; scale 1.000x)



SELECTED v0.3



Residual: CPU 4.81 ms (0.04%); NPU 5.06 ms (0.02%). Bars: arithmetic means; speedups: prompt GMs.

Replay and proposal share one timed call. Phase-1 steady-state throughput excludes the full-prefix state cost.

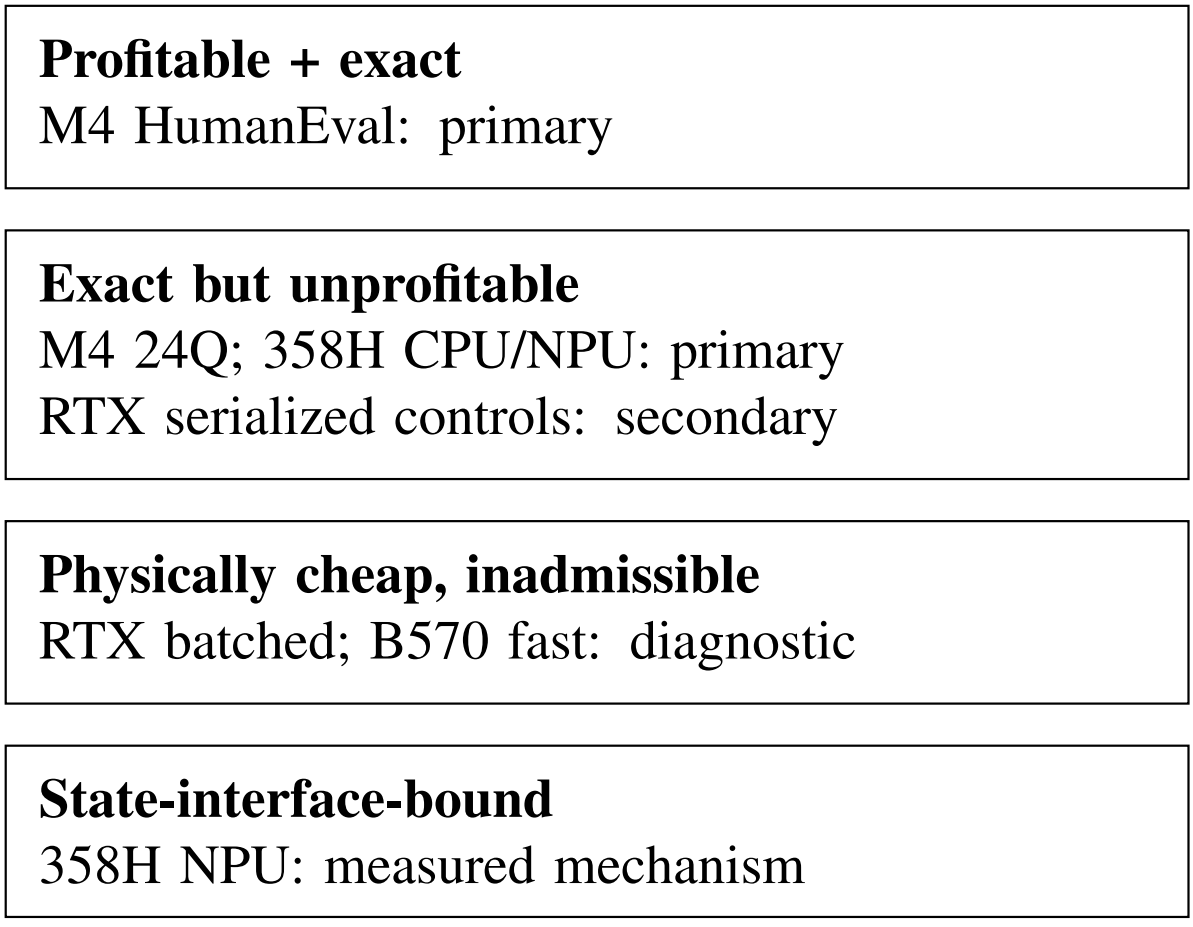
Speedup 95% CIs: CPU [0.693, 0.771]; NPU [0.333, 0.369]. Initial prompt prefill is excluded from Phase 3.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

Figure 5: OLD / PRIOR candidate v0.2

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 2.077x)



PRIOR candidate v0.2

Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

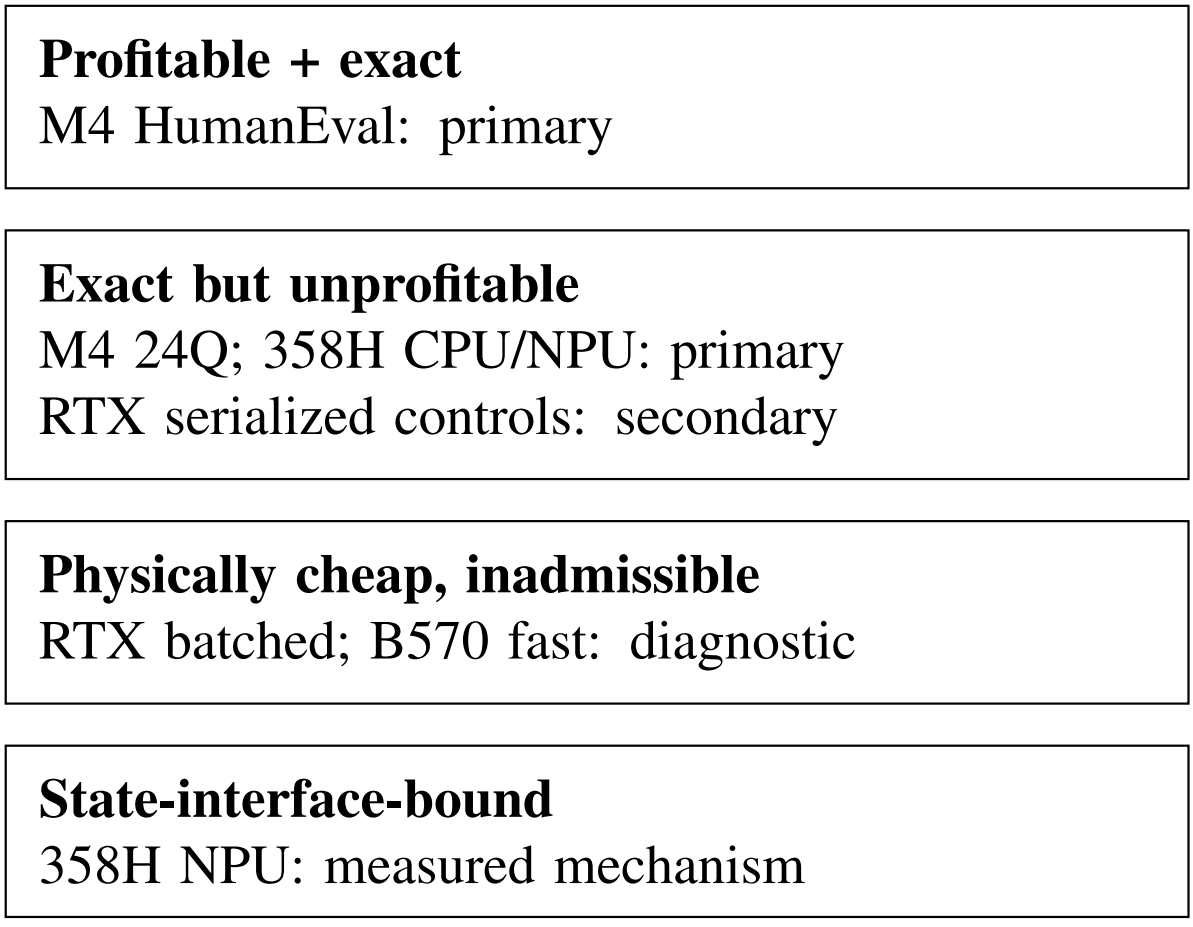
P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.

Figure 5: OLD / SELECTED v0.3 (appendix A1)

Both panels 6.75 in wide. Print at 100%; do not fit to page.

OLD manuscript v0.1 (native 3.25 in; scale 2.077x)



SELECTED v0.3 (appendix A1)

Categorical regimes under the frozen correctness contract

Mechanism	Not admissible	Correctness-qualified	
	No admissible speed claim	Negative ($S < 1$)	Profitable ($S > 1$)
Workload conditioning		● **M4 representative 24Q [P]** 72/72 exact	● **M4 HumanEval30 [P]** 90/90 exact
Verification constraint	○ **RTX batched [M]** Cheap micro path; gate fails	● **RTX serialized [S]** CPU→RTX / RTX→RTX; exact	
Correctness barrier	○ B570 fast: 45/90 [D] ○ B570 serialized: 60/90 [D]		
Target-side cost		● **358H CPU [P/M]** 90/90; target side 87.34%	
State-interface cost		● **358H NPU [P/M]** 90/90; replay + proposal 58.35%	

P primary · S secondary · M mechanism · D diagnostic. Fractions count exact token-ID pairs.

Exact checkpoint artwork; no reconstructed values. Captions assessed in FIGURE_REVIEW.md.